

Entities, relationships, and big concepts

The full structure the system extracted and the GraphRAG community detection produced, from the current run. Every entity is grounded in a verbatim quote from a source paper.

180

Canonical entities (concepts)

439

Typed causal relations

8

Big concepts (Leiden communities)

1 Big concepts (GraphRAG community reports)

Entities that recur together across studies are clustered by Leiden community detection; each cluster is summarised into one higher-level concept. This is the emergent, data-driven layer — not a predefined list.

Mechanism_Response 39 MEMBERS · 16 STUDIES

Structured reasoning uptake

This cluster captures the learner-side response to reasoning aids, examples, feedback, and virtual cases: some supports are adopted and processed, while others are ignored, judged too complex, or seen as offering limited added value. In the realist logic, the mechanism is the learner's appraisal and use of the offered structure or feedback, which can lead to improved diagnostic estimation, recognition of missing information, repeated correct answers, or illness-script development when the resource is taken up. Where the support feels time-consuming, overly complex, or insufficiently useful, the response is non-use or only limited learning.

Members include: subtle limited added value of SE models, virtual experience of real-time reasoning techniques, error identification and correct-solution comparison, complex individual decision-making situations, learning after failed retrieval, reflection on and improvement of reasoning, familiarity with existing structured tools, time-compressed virtual patient cases

Mechanism_Resource 29 MEMBERS · 22 STUDIES

Contrastive Diagnostic Scaffolding

This cluster represents a family of teaching resources that structure diagnosis around contrasts: compare-and-contrast prompts, differential tables, feature-oriented instructions, paired examples, and guided combined-strategy practice. In realist terms, these resources work by offering a schema for noticing shared and distinguishing features, which can reorganise knowledge and shift learners toward more structured, combined reasoning; in novices this is associated with improved performance and greater accuracy after contrastive learning. The cluster also includes resources that can create confusion when combined instructions are too dense, showing that the mechanism depends on how the instructional input is framed and used.

Members include: typed feature list with terminology explanation, combined-strategy accuracy hypothesis not supported, analytic cytopathology feature training, novice diagnosticians, increased use of schema-based reasoning, consideration of alternatives and conflicting findings, traditional cardiac physical diagnosis teaching, differential diagnosis evidence table

Intervention 25 MEMBERS · 12 STUDIES

Reasoning Practice Scaffolding

This cluster represents interventions that make clinical reasoning visible and actionable: students are prompted to query an initial hypothesis, compare it with contradictory or absent findings, and practice reasoning in individual, small-group, seminar, or course-based formats. The resources include explicit modelling/explanation of expert reasoning, teacher-led consultation, video-enriched or virtual patient cases, and bias-focused prompts such as search-satisficing instruction and self-monitoring. In realist terms, these interventions offer structured cues and practice opportunities that can trigger learners to think during hypothesis query, understand physician thought processes, and refine or correct diagnoses.

Members include: CFS case-based seminar, second-diagnosis search, misleading feature or diagnosis suggestion, explicit modelling/explanation of expert reasoning, student engagement in clinical reasoning practice, querying initial diagnostic hypothesis, oversearching for nonexistent second diagnosis, three-consultation clinical reasoning skill framework

Mechanism_Response 23 MEMBERS · 18 STUDIES

Structured Diagnostic Knowledge Building

This cluster captures a mechanism-response in which learners move from poorly developed illness scripts and weak links to disease mechanisms toward more structured diagnostic knowledge. Worked examples, case-based reflection, self-explanation, and embedded prompts appear to support selective organisation, storage and retrieval of knowledge, detection of misunderstandings, and proposed illness-script enrichment, often with lower cognitive load and greater confidence. The same response can also be seen in delayed-test gains and improved MCQ performance, especially when learners engage diligently with practice and feedback.

Members include: students lacking well-developed illness scripts, logical connection to disease mechanisms, structured diagnostic process schema, natural expert non-analytic abstract reasoning, proposed illness-script enrichment, video-game-experienced surgical learners, lower cognitive load during example study, structured reflection diagnostic case practice

Mechanism_Response 18 MEMBERS · 8 STUDIES

Pressure-driven case patterning

This cluster captures a response pattern in which learners confronted with simulated ward multitasking, short case windows, multiplied patient encounters, and competing pager demands move toward quicker recognition and more secure case representations. Across the studies, the offered resources include varied cases, case-based diagnosis/explanation, and theory-provided tutorials or games, but the learner response is often frustration, limited perceived benefit from theory, or no discernable difference in approach. The mechanism is the shift from deliberate theoretical processing to implicit learning and faster recognition under pressure, especially for students near transition to junior doctor.

Members include: tutorial-provided theoretical understanding, students near transition to junior doctor, limited perceived benefit from theory, simulated multi-tasking ward experience, varied same-process clinical cases, diagnose explain and learn from cases, implicit learning and faster recognition, difficulty too high for some students

Mechanism_Response 17 MEMBERS · 14 STUDIES

Unguided Diagnostic Reasoning

This cluster captures a response pattern in which learners are left to diagnose, notice features, or reason spontaneously with little or no explicit learning process guidance. Across the studies, this shows up as minimal exposure, speeded single-diagnosis work, natural-strategy or spontaneous-reasoning instruction, and unguided practice with biasing or incorrect suggestions; the common outcome is weaker or less effective implementation, shorter engagement, missed self-explanation, and lower accuracy. In realist terms, the mechanism-response is the learner's unguided diagnostic processing, which is especially visible when novices or low-support conditions lead them to rely on whatever strategy they already have rather than a newly scaffolded one.

Members include: self-generated helpful abnormal-cell features, minimal exposure to cytopathology images, no explicit learning processes, speeded single-diagnosis task, simply giving a diagnosis, lower accuracy under incorrect suggestion without guidance, missed additional self-explanation, natural-strategy instruction

Mechanism_Response 17 MEMBERS · 15 STUDIES

Guided Self-Explanation

This cluster represents a mechanism-response pattern in which learners are given prompts, explicit cognitive-feature focus, or guided consultation steps and then respond by generating explanations, thinking more actively, and keeping their reasoning on track. Across the studies, this response is associated with knowledge elaboration and reorganisation, inference generation, and accelerated acquisition or transfer of targeted reasoning skills, especially when working through written clinical cases with limited prior familiarity. In realist terms, the offered resources appear to elicit a more productive reasoning process that supports diagnostic case solving and consultation performance.

Members include: inference generation and knowledge integration, self-explanation diagnostic case solving, limited familiarity with clinical topic, case reading and self-explanation prompt without feedback, guided performance of consultation reasoning steps, intended insight into cognitive features of reasoning, student performance consultation, written jaundice clinical cases

Outcome 12 MEMBERS · 13 STUDIES

Error-Resistant Diagnostic Learning

This cluster describes an outcome where learners become less likely to miss the correct diagnosis or be led astray by misleading information, even though their pathophysiology knowledge and deep conceptual understanding do not substantially improve. It appears in settings with difficult far-transfer cases, worked examples containing errors plus KCR feedback, and official diagnosis feedback after each case, alongside unsupported similar-case reasoning with reduced feedback. In realist terms, the offered resources and case demands seem to produce a more error-resistant diagnostic response, while leaving the deeper knowledge base largely unchanged.

Members include: official diagnosis feedback after each case, difficult far-transfer diagnostic cases, insufficient biomedical integration in illness scripts, insufficient deep conceptual understanding, incomplete error-prone prior knowledge, worked examples with errors and KCR feedback, unawareness or incapacity to solve errors, resistance to misleading information

2 Entities, by realist type

All canonical concepts, with the number of studies each appears in and one representative source quote. Entities are the nodes of the knowledge graph.

Context **18 concepts** — Pre-existing conditions of the learner, group, or setting.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
non-expert surgical learners	11	this study was performed among fourth-year medical students just before attaining the license of medical doctor.
novices not using nonanalytic reasoning	7	Therefore, they are not yet likely to have well-developed illness scripts.
advanced students with prior endocrinology training	5	A sample of 48 undergraduate psychology students enrolled at McMaster University participated in this 3-phase experiment for cours...
potential lack of clinical exposure	5	Given the potential lack of exposure to some relevant aspects of clinical medicine during small-group and face-to-face teaching, a...
prior exposure to specific ECGs	5	This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reaso...
realistic unrestricted learning conditions	5	Of these, 15 occurred in two e-seminars, which allowed for repeated testing of these 'intervention items'.
availability-bias cases with plausible common diagnosis	4	The unique diagnostic approaches used by GPs tend not to be explicitly taught to medical students.
case complexity beyond students' knowledge	4	Especially when learners are confronted with complex tasks, elaborated feedback which combines these kinds of information (in comp...
clerks facing unfamiliar jaundice cases	4	students' prior exposure to similar clinical problems in the undergraduate problem-based learning curriculum. Heart failure was id...
absence of active-processing prompts	3	Students found that existing tools, with which they were familiar, were very useful and helped to reduce their stress levels.
clinically different expertise levels	3	Scores were compared between groups of different levels of expertise: surgeons (licensed surgeon), residents (surgical trainee), i...
incomplete error-prone prior knowledge	3	Self-explanations are not necessarily formal and complete, but may be fragmented, incorrect or incomplete; reflecting, in fact, wha...
lack of ward exposure	3	Follow-up of and reflection on patients is a powerful discipline for the learning and construction of clinical reasoning skills (K...
search-satisficing cases with two diagnoses	3	SSB consisted of one case using an x-ray and one using an ECG. Each case had two correct diagnoses.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
cascade of data abnormalities	2	Readers of Clinics in Laboratory Medicine no doubt share an interest in laboratory science and clinical pathology, but it is likel...
confusable ECG categories with shared features	1	similar by virtue of their having features in common, and 2 known to be easily confusable based on the results of previous studies...
non-textbook diagnostic courses	1	Therefore, cases do not necessarily have to present “textbook” solutions, treatments, or behaviours to encoun- tered problems.
video-game-experienced surgical learners	1	The majority of surgical residents and surgeons have experience with video games and are therefore likely to embrace using a video...

Intervention **36 concepts** — The educational methods and resources studied.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
case-based clinical reasoning seminar intervention	8	This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reaso...
self-explanation diagnostic case solving	4	They were then asked to read each case while self-explaining aloud for 8 minutes and then they had 2 minutes to write down their d...
cognitive theory and interactive cases	3	The tutorial covered cognitive theory (dual process theory and cognitive and affective biases), prioritisation and when to ask for...
analytic cytopathology feature training	2	they were taught the morphological features of normal and abnormal cells by an experienced cytology trainer and shown 20 pairs of ...
Casebook virtual patient application	2	An iPad-based Virtual Patient interactive software system was developed entitled Casebook.
combined self-explanation and SE model observation	2	Participants of the peer-SE and expert-SE groups were then instructed to listen to an 8-min recording of either peer or expert SE
diagnostic decision tree VP cases	2	Diagnostic decision tree software can be used to create VP cases that give virtual experience of these real-time clinical reasonin...
differential diagnosis case practice	2	In the differential diagnosis condition, participants were requested to read the case and write down the most likely diagnosis for...
structured reflection diagnostic case practice	2	structured reflection compared with the generation of immediate or differential diagnosis while practising with clinical cases
AM I HERO NOW tool introduction	1	Therefore, a decision making tool (AM I HERO NOW) was introduced to students, to be used when faced with making a decision
case-based worked examples with feedback variations	1	a case-based, worked example approach was realised in a computer-based learning environment with the intention of facilitating med...
CFS case-based seminar	1	The educational intervention consisted of a 90-minute, interactive, case-based seminar of 15 students and one instructor following...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
comprehension-stage teacher-led consultation	1	following the demonstration and deconstruction phase, the next consultation should again be performed by the teaching clinician
computer-assisted key feature testing	1	Testing was spaced across weeks and students were given immediate educational feedback on their answers.
contrastive learning instruction	1	The remaining participants were explicitly instructed to draw comparisons between the diagnostic category being learned and anothe...
diligent study and practice	1	with expertise developed over years of diligent study and practice, become increasingly adept at learning to trace back
final combined-strategy testing phase	1	Prior to a final test (test 3), participants in both groups were instructed to adopt a combined analytic/nonanalytic diagnostic st...
Integrated Cases Section course	1	Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students
one-to-one ECG diagnostic training	1	all participants were taught general information regarding the 12 leads in an ECG during one-to-one teaching sessions with the exp...
serious game for biliary decision-making	1	A serious game was developed to train surgical residents in clinical decision-making regarding biliary tract disease.
simulated multi-tasking ward experience	1	Simulated individual multi-tasking experiences have the potential to make students more prepared for work.
simulator session with SBAR beside phones	1	During the simulator session, the SBAR tool was written down next to the phones.
speeded single-diagnosis task	1	Both accuracy and speed were stressed, and participants were instructed to move on to the next page to solve a word puzzle after t...
spontaneous reasoning instruction	1	whereas the other half were given no instructions regarding how best to approach the diagnosis of test cases (i.e. the spontaneous...
student performance consultation	1	in the third consultation, the student follows the model outlined in the first and second consultations
tacit pick-up approach to clinical reasoning	1	clinical reasoning should not be considered as something tacit that students simply 'pick up as they go along'

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
teaching clinical reasoning to students	1	teaching clinical reasoning also provides an excellent opportunity for clinicians to consider and review their own clinical reason...
three-consultation clinical reasoning skill framework	1	An adaptation of this can be used as a model to teaching the skill of clinical reasoning across three consultations.
time-compressed virtual patient cases	1	VP cases can be used to compress time, allowing students to observe strategies such as ' test of time ' o r ' test of treatment ' w...
traditional cardiac physical diagnosis teaching	1	traditional teaching of cardiac physical diagnosis involves diagnosis-specific lists of features, structured around steps in the ca...
uncertain pedagogy for decision-making theory	1	A significant difficulty with delivering a tutorial is the uncertainty from the literature on how students should be taught to mak...
unguided diagnostic practice with biasing feature	1	The other half of the participants were not given instructions that would promote any particular reasoning process.
unsupported similar-case reasoning with reduced feedback	1	what medical students learn spontaneously from reasoning through a series of cases that have a similar underlying biomedical probl...
video-enriched virtual patient cases	1	The aim of our video enriched VP cases is to simulate GP consultations revolving around these core presentations, giving students ...
worked examples with errors and elaborated feedback	1	In order to enhance the effectiveness of the approach, two additional measures were implemented: erroneous examples and elaborated...
worked examples with errors and KCR feedback	1	In the condition /C212errors and KCR feedback/C213the learners had to deduce the right step in the diagnostic process from the sub...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
multiple tasks and acutely unwell patients	5	During their simulation session, the students were faced with multiple tasks and acutely unwell patients, and therefore had to pri...
opportunities to link biomedical and clinical knowledge	5	Self-explanation is a constructive learning strategy that consists of generating explanations to oneself in response to learning m...
background information without highlighted tested aspects	4	This compression of time also allows the evolution of chronic disease, longitudinal management and treatment effects to be illustr...
combined analytic-nonanalytic strategy instruction	4	New ECGs often look like ECGs that have been seen before (i.e. during training). Trust this sense of familiarity, but realise that...
natural-strategy instruction	4	/C212Assign a diagnosis to each ECG using whatever strategy comes naturally to you. Feel free to use the feature list, but approach...
organised records with error discussions	4	In the condition with errors, the student makes several severe errors. The choice of errors was guided by the error taxonomy descr...
practice with ECG examples	4	For each diagnostic category, participants were presented with 4 examples. For the first 2 ECGs in each category, the instructor i...
alternative diagnoses prompt	3	After that, they were asked to think about alternative diagnoses if their initial hypothesis were to prove incorrect, write them d...
correct and incorrect answer explanations	3	Feedback for intervention items contained all options from the key feature long menu that were considered 'correct'. It also conta...
diagnosis-specific table of clinical features	3	students in the analytic reasoning group received an answer sheet containing a table designed to enhance analytic reasoning.
elaborated resident explanations of clinical reasoning	3	selected the example that provided the most elaborate explanations for the clinical findings and differential diagnosis.
embedded prompts	3	A total of 12 prompts, six of each type, were incorporated into the four resident SE recordings.
official diagnosis feedback after each case	3	After the students had completed a case, the official diagnosis was given, as they would have received similar feedback if they had...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
clinical case practice with feedback	2	Students then individually practiced CFS using given clinical cases by developing a diagnosis and treatment plan for each case. Fe...
step-by-step matching to alternative diagnoses	2	The additional schema allows the steps of the diagnostic process to be contextualised by providing structured information on all s...
additional circumspection demand	1	Further encouragement to engage in additional circumspection (particularly in a novice, where, by definition, more analytic than n...
automated unarticulated clinician reasoning	1	The learner initially observes a consultation by the teaching clinician, followed by the teacher explaining the reasoning processe...
case reading and self-explanation prompt without feedback	1	participants were asked to read each case and to generate self-explanations out loud. They received no feedback on the quality of ...
clinical reasoning-oriented course exposure	1	Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students
compare-and-contrast category prompts	1	participants were asked to compare and contrast the features present on each example with those of a normal ECG and with those of ...
complex individual decision-making situations	1	experiencing complex, individual decision making situations through simulation has the potential to make students more prepared fo...
diagnostic schema framework grouped by clinical features	1	The schema framework (Fig. 2) grouped diagnoses according to clinical features based on murmur location and timing (e.g. 'apex bea...
explicit cognitive-features focus	1	This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reaso...
incorrect diagnostic suggestion	1	Six of the ECGs were presented with no additional information, 6 were presented with the correct diagnosis, and 6 were presented w...
instructional input	1	The relative reliance on analytic and similarity-based processes is amenable to instruction and dependent on expertise.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
minimal exposure to cytopathology images	1	Our study clearly illustrates the extent to which inexperienced observers are able to generate diagnostic criteria without assista...
minimised learner reflection	1	This procedure was used to minimise the likelihood that participants would engage in reflection in the other experimental condition...
mnemonic decision-making tool	1	The tool was designed to be a memorable mnemonic, incorporating multiple aspects of clinical decision making, with the aim of prom...
natural expert non-analytic abstract reasoning	1	Experts use fast, largely non-analytic reasoning, tend to skip steps and to reason at a higher level of abstraction (Bordage 2007)
nonannotated normal-abnormal image pairs	1	no feature-driven training was provided, and the only image annotation used during training was restricted to the labels "normal" ...
peer or expert self-explanation audio models	1	As peer and expert SE examples, we used audio-recorded SE. To reflect the variety of examples possibly encountered in the clinical...
readily available SBAR structure	1	During the simulator session, the SBAR tool was written down next to the phones. Making it readily available increased the use of ...
search-satisficing bias instruction	1	Students were taught two biases: search satisficing and availability. With SSB, the clinician may identify one diagnosis but miss ...
simple diagnosis answer sheet lacking table	1	Students in the control group were simply asked to provide one most probable diagnosis on an answer sheet that lacked the table pr...
ten-second case time limit	1	The player had a maximum of 10 s to solve each case.
tutorial-provided theoretical understanding	1	A tutorial session was introduced providing a theoretical understanding of the process of clinical decision making.
typed feature list with terminology explanation	1	Participants in this group were also given a typed list of features for future reference, and a full explanation of the relevant c...
varied same-process clinical cases	1	We developed two series of five cases with the same underlying pathophysiological process but with different presentations and caus...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
virtual experience of real-time reasoning techniques	1	Diagnostic decision tree software can be used to create VP cases that give virtual experience of these real-time clinical reasonin...
virtual prior experience with real patient videos	1	We chose to use real patient videos as the basis of our VP cases to provide a virtual 'prior experience' to enable these strategie...
worked examples of diagnostic cases	1	The learning environment, integrated in the CASUS learning tool, consisted of a schema illustrating the steps of the diagnostic pr...
written jaundice clinical cases	1	A set of four cases, one per category, was used in the training phase (training cases).

Mechanism Response

55 concepts — How learners cognitively or emotionally react.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
access to problem-solving knowledge	9	our data are likely to reflect the true effect of repeated testing on the ability to access the knowledge needed to solve the clini...
attention to schema-based distinguishing features	5	Completing the table presented on the answer sheet forced the consideration of alternative diagnoses and of findings that conflict...
relative reliance on analytic and similarity-based processes	5	participants undertook test 2, for which they were asked to provide their first impression of each image based on their similarity...
cognitive load overwhelm	4	may simply overwhelm an individual's cognitive load without any effect on outcomes.
knowledge elaboration and reorganisation	4	This process of elaboration on the case information and search for meaning helps to activate additional relevant knowledge and fos...
active tracking of demonstrated reasoning	3	even incorrect or erroneous self-explanations may promote learning if the learner, detecting these misunderstandings, then tries t...
appreciation of real-patient VP videos	3	they are provided the opportunity to follow the patient 's course of disease or illness, and thereby witness the effects of manage...
awareness of knowledge limits	3	learners become more aware of what they know or understand well, but are also confronted with uncertainties, forgotten information...
common incorrect diagnosis selection	3	Therefore, learners must identify the error and understand the correct solution by systematically comparing the incorrect solution...
easy comparison with near-peer reasoning	3	which allows the latter to relate to it easily and to appreciate similarities and differences.
sole feature-based analysis	3	participants were instructed to base their diagnosis solely on an analysis of the cytological features observed.
inference generation and knowledge integration	2	When solving less familiar cases, students generated significantly more biomedical inferences than with familiar cases.
logical connection to disease mechanisms	2	This finding further reinforces the importance of stimulating students to reactivate underlying pathophysiological mechanisms when ...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
lower cognitive load during example study	2	studying worked examples imposes lower levels of cognitive load on the learner than solving training problems, particularly becaus...
reflection only to some degree	2	Generating alternative diagnoses is likely to have led the students in the differential diagnosis group to also engage in reflectio...
schema construction investment	2	Therefore, more cognitive resources can be invested in processes of schema construction.
stimulated clinical diagnostic reasoning	2	patient video cases were found to ' strongly stimulate clinical diagnostic reasoning processes ' (Balslev et al. 2010).
testing and readjusting hypotheses	2	Gameplay closely mimics the problem-solving ability required in diagnostic and therapeutic reasoning (i.e., testing and readjustin...
understanding expert thinking and reasoning	2	this is to assist the student to understand what the clinician was thinking and how they were reasoning.
greater analytic-tendency reliance	1	the aim of the feature manipulation was to induce people towards placing greater reliance on their analytic tendencies
increased use of schema-based reasoning	1	schema-instructed students were more likely to report features from the schema in arriving at their diagnosis. This suggests a hig...
insufficient deep conceptual understanding	1	For most students, the information that a specific diagnostic conclusion or procedure is not adequate seems not to be sufficient to ...
intended insight into cognitive features of reasoning	1	This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reaso...
knowledge organisation into structured knowledge	1	The results on the written test provide evidence that schemas modified knowledge organisation by increasing the amount of structure...
learning after failed retrieval	1	33.3% of students who failed to provide a correct answer in one test went on to submit a correct answer in a subsequent test of th...
limited contradictory-and-absent finding checking	1	The latter does not necessarily happen during the generation of differential diagnoses, but it seems to be a crucial aspect of the...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
limited perceived benefit from theory	1	SE5 – The clinical based stuff you were talking about was good, but the sort of theory behind it, I don't know if it would benefit...
little learning from same-level examples	1	participants at the exact same level may have little to learn from such examples.
missed additional self-explanation	1	Justification and mental model revision prompts presumably engaged students in additional SE while they listened to the example
more active thinking during self-explanation	1	Unfamiliarity seems to act as a trigger for students to engage in more active thinking; the greater diagnostic challenge encourage...
more secure faster case representations	1	They built up representations faster, became more secure, used more specific language, and tested their knowledge against case info...
no explicit learning processes	1	As many students did not show any explicit learning processes, this result raises doubts about the students' ability to take full ...
observing test-of-time and treatment strategies	1	allowing students to observe strategies such as ' test of time ' o r ' test of treatment ' which GPs may use to define their diagno...
oversearching for nonexistent second diagnosis	1	Table 3. Performance on false positive cases Group Correctly identified diagnosis (%) Identified a second (nonexistent) diagnosis ...
pattern recognition from multiple-case experience	1	This experience builds pattern recognition, meaning the more experience a student has, the more likely they will make a correct di...
perceived excessive time demand	1	SC1 – You don't have time to sit there and think about every single question on that piece of paper.
perceived tool complexity	1	SD3 – I thought there was a lot of information on it. Maybe a bit too much.
perceived use of clinical experience	1	In total, 60.6 % stated that decisions in the serious game are based on realistic elements, and 66.7 % found their clinical experi...
possible confusion from combined instructions	1	An alternative explanation for the relatively poor performance in the final test, which occurred in both groups of participants, i...
practising and refining GP reasoning	1	The VP cases give students opportunities to practise and refine unique clinical reasoning skills typically deployed by general pra...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
processing elaborated feedback information	1	coping with erroneous examples and processing elaborated feedback information (or compensating superficial feedback by self-explana...
proposed illness-script enrichment	1	reflection enriched the illness scripts they had already stored, which, in turn, facilitated the diagnostic process when they encou...
reasoning kept on track	1	if the student's reasoning is unclear, the clinical teacher can interject with questions such as 'Why did you ask that?' or 'How d...
recognition of inadequate information gathering	1	SD3 – I should probably gather more information before I decide who needs to be seen. I didn't ask for observations or anything wh...
reduced fear during senior referral	1	SB2 – The SBAR structure was really useful. It took a bit of the fear away definitely
reflection on and improvement of reasoning	1	the benefits of reflecting upon and improving one's own reasoning may include:
reminded of previous cases	1	Many students (n = 25) stated in their think-aloud protocols that they had been reminded of a previous case in the series.
second-diagnosis search	1	For the SSB cases, 52% of students with CFS training and 48% in the control group initiated a search for the second diagnosis
self-generated helpful abnormal-cell features	1	Table 2 summarizes the features that participants in the nonanalytic group claimed were helpful in identifying abnormal cells, tog...
simply giving a diagnosis	1	when they simply give a diagnosis or even when they generate a differential diagnosis, as they are usually recommended to do durin...
storage and retrieval of knowledge	1	the observed score difference can be attributed to an effect of repeated testing on the storage and retrieval of knowledge rather ...
stress from competing patient and pager demands	1	SA3 – It was stressful. When you have a sick patient and then the page goes off, that was stressful.
student frustration	1	Of the students, 8 of 17 found the game frustrating (compared to 1 of 11 trainees and 1 of 6 experts).

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
unawareness or incapacity to solve errors	1	The remaining 14 participants were either not aware that they made errors of reasoning or were incapable of solving them, as the p...
unguided or less effective strategy implementation	1	we cannot make conclusive statements about whether participants in the spontaneous reasoning condition did or did not implement a ...

Outcome

29 concepts — The reasoning and learning results.

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
increased learning time	13	in our study, elaborated feedback resulted in a substantial increase in learning time.
improved diagnostic accuracy	7	Mean diagnostic accuracy scores were significantly higher in the analytic reasoning group than in the control group
accurate conclusions about case variations	6	Participants in the contrastive learning condition performed significantly better than those in the non-contrastive learning condi...
perceived usefulness for developing consulting skills	5	Thus the added value of SE models was altogether subtle, only evidenced on training cases, and expert SE models were not more effe...
reduced ability to exploit cases	5	this result raises doubts about the students' ability to take full advantage of learning from cases.
accelerated acquisition of targeted reasoning skills	4	This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reaso...
better diagnostic performance one week later	4	The reflection condition was the only condition to show that learning had occurred when comparing performance in the learning phase...
decline in final-test discrimination	4	When only KCR feedback is given, erroneous examples are detrimental.
intended acquisition of primary-care reasoning skills	4	Follow-up of and reflection on patients is a powerful discipline for the learning and construction of clinical reasoning skills (K...
reduced accuracy with biasing feature	4	After 1 week, students who had been instructed to reflect during the learning phase outperformed students in the other conditions o...
expert models no better than peer models	3	Students in the condition /C212with errors and elaborated feedback /C213 achieved the highest scores in diagnostic knowledge, foll...
improved retained structured knowledge	3	Students in the schema group performed better on structured knowledge questions (62% versus 42%; $p = 0.001$), but not on factual kn...
incorrect second diagnoses without difference	3	Sixty-four percent of the CFS group and 77% of the control group incorrectly identified a second diagnosis. There were no signific...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
objective prioritisation learning	3	Making it readily available increased the use of SBAR. The students felt more comfortable using the structured, SBAR approach
partial remediation for some ignored errors for others	3	More importantly, 33.3% of students who failed to provide a correct answer in one test went on to submit a correct answer in a sub...
pathophysiology knowledge did not really improve	3	Since one of the probes for the learner report invited students to report on what they knew now about the pathophysiology of the d...
primary-care-specific reasoning skill acquisition	3	a teaching approach using this framework is proposed to assist in the development of clinical reasoning ability.
barrier to learning clinical reasoning	2	According to the “flow theory,” the level of difficulty is too high for some students, which could be adjusted by incorporating di...
criteria similar to standard cytopathology features	2	Although written in layman’s terms, the morphologic descriptions are surprisingly similar to the standard list of features provide...
effective and efficient worked-example approach	2	Our results demonstrate that the case-based, worked example approach is effective and efficient.
faster nonanalytic responses	2	Speed of response to test images was generally faster under nonanalytic than under analytic conditions.
insufficient biomedical integration in illness scripts	2	unsupported case analyses are in no way sufficient for integrating biomedical knowledge in illness scripts.
AM I HERO NOW non-use	1	the ‘AM I HERO NOW’ tool, although introduced, was not actually used by students as a decision making tool during their simulated ...
combined-strategy accuracy hypothesis not supported	1	The hypothesis that a combined nonanalytic/analytic reasoning strategy can achieve diagnostic accuracy greater than that employing...
guided performance of consultation reasoning steps	1	by performing the history, physical examination and directing investigations, explaining to the clinical teacher their reasoning a...
higher arterial hypertension MCQ scores after six weeks	1	experimental group students achieved 5.59 of 7 points (SD = 0.95), whereas control group students achieved only 5.07 points
illness-script development for diagnosis	1	pools of similar cases are required so that students can properly develop their illness scripts in order to make suitable and vali...

CONCEPT	STUDIES	REPRESENTATIVE QUOTE
incomplete patient assessment after interruption	1	When I was going to the patient I was paged, so I forgot to finish the ABCDE. I didn't really assess the patient properly.
rectified incorrect diagnosis	1	querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect di...

3 Relationships

439 directed, typed causal links. Distribution: LEADS_TO 118 · TRIGGERS 113 · PROVIDES 105 · ENABLES 79 · CONSTRAINS 22. A sample of the links follows; the full set is in the interactive console and the canonical data files.

SUBJECT	PREDICATE	OBJECT
Cont limited follow-up in short rotations	CONSTRAINS	Outc learning and construction of reasoning skills
Cont students lacking well-developed illness scripts	CONSTRAINS	Outc less effective learning than structured reflection
Cont absence of active-processing prompts	CONSTRAINS	Outc no transfer-case added value of SE models
Cont novice senior medical students	CONSTRAINS	Outc no significant second-diagnosis improvement
Cont short undergraduate primary care rotations	CONSTRAINS	Outc accelerated acquisition of targeted reasoning skills
Cont familiar clinical topic	CONSTRAINS	Outc no demonstrated benefit for familiar cases
Cont case complexity beyond students' knowledge	CONSTRAINS	Outc reduced helpfulness of analytic reasoning
Cont novices not using nonanalytic reasoning	CONSTRAINS	Outc failed reduction of diagnostic error by novices
Cont newly trained ECG novices	CONSTRAINS	Outc lower accuracy than combined reasoning instruction
Cont diagnostic expertise level	CONSTRAINS	Outc modifiable expertise-dependent processing reliance
Cont GP diagnostic approaches not explicitly taught	CONSTRAINS	Outc primary-care-specific reasoning skill acquisition
Cont unknown systolic murmur requiring information integration	CONSTRAINS	Outc lower practical diagnostic accuracy than schema group
Cont misleading feature or diagnosis suggestion	CONSTRAINS	Outc resistance to misleading information
Cont incomplete error-prone prior knowledge	CONSTRAINS	Outc insufficient biomedical integration in illness scripts
Cont very poor prior knowledge	CONSTRAINS	Outc insufficient biomedical integration in illness scripts
Cont typical clinical presentation	CONSTRAINS	Outc similar acute pyelonephritis accuracy
Cont lack of ward exposure	CONSTRAINS	Outc illness-script development for diagnosis

SUBJECT	PREDICATE	OBJECT
Cont novice senior medical students	CONSTRAINS	Outc limited uncommon-diagnosis recognition
Cont difficult far-transfer diagnostic cases	CONSTRAINS	Outc no far-transfer diagnostic accuracy advantage over control
Cont difficult far-transfer diagnostic cases	CONSTRAINS	Outc no far-transfer performance advantage over control
Cont complex learning environments	CONSTRAINS	Outc poor diagnostic knowledge performance
Cont well-defined illness scripts for familiar problems	CONSTRAINS	Outc no diagnostic benefit for familiar topics
Cont students lacking well-developed illness scripts	ENABLES	Mech high cognitive load and initial confusion
Cont students lacking well-developed illness scripts	ENABLES	Mech knowledge elaboration and reorganisation
Cont usual differential-diagnosis teaching norm	ENABLES	Mech reflection only to some degree
Cont diagnostic expertise level	ENABLES	Mech relative reliance on analytic and similarity-based processes
Cont incorrect initial diagnosis	ENABLES	Mech thinking during hypothesis query
Cont unfamiliarity with pagers	ENABLES	Mech stress from competing patient and pager demands
Cont students with deficient diagnostic knowledge base	ENABLES	Mech schema construction investment
Cont students lacking well-developed illness scripts	ENABLES	Mech proposed illness-script enrichment
Cont novice diagnosticians	ENABLES	Mech greater non-analytic tendency reliance
Cont no prior use of AM I HERO NOW	ENABLES	Mech perceived tool complexity
Cont realistic unrestricted learning conditions	ENABLES	Outc increased learning time
Cont novice diagnosticians	ENABLES	Mech avoidance of single-mode reasoning
Cont correct initial diagnosis	ENABLES	Outc no harm to correct diagnosis
Cont no prior prioritisation training	ENABLES	Mech stress from competing patient and pager demands

SUBJECT	PREDICATE	OBJECT
Cont well-defined illness scripts for familiar problems	ENABLES	Mech limited room for elaboration and reconstruction
Cont limited prior decision responsibility in shadowing	ENABLES	Mech recognition of inadequate information gathering
Cont newly trained ECG novices	ENABLES	Mech greater analytic-tendency reliance
Cont prior exposure to specific ECGs	ENABLES	Outc higher accuracy for previously seen ECGs
Cont newly trained ECG novices	ENABLES	Mech unguided or less effective strategy implementation
Cont limited familiarity with clinical topic	ENABLES	Outc better diagnostic performance on less familiar cases
Cont less developed illness scripts in unfamiliar area	ENABLES	Mech inference generation and knowledge integration
Cont prior exposure to specific ECGs	ENABLES	Mech similarity-based processes operational
Cont familiarity with existing structured tools	ENABLES	Mech greater comfort making senior referral

Generated live from the current run's database. Entities carry a verbatim quote resolved to a character span in the source; relations that violate the ontology's domain/range are demoted, not shown here. The full graph is interactive in the review console (Knowledge graph tab).